FROM LIPID MEMBRANES TO LIPID NANOPARTICLE DESIGN WITH MARTINI 3 MODEL

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ABSTRACT

Lipid assemblies exhibit complex phase behavior that governs organization, remodeling, fusion, and molecular transport. Understanding these collective phenomena is essential for deciphering lipid-mediated biological processes and designing lipid-based delivery systems. Experimental approaches often lack the resolution needed to reveal the underlying molecular mechanisms, while atomistic molecular dynamics simulations remain computationally demanding [1]. Coarse-grained (CG) molecular dynamics simulations, particularly with the Martini force field, provide an attractive alternative [2]. In this talk, I will present recent advances in the Martini 3 lipid models that improve membrane properties and phase behavior [3,4]. Building on refined lipid-tail parameters and an expanded lipid library, these developments reproduce phase transitions while capturing lipid domain formation, ion-mediated phase separation, and non-lamellar phases. These advances naturally extend to lipid nanoparticles (LNPs), enabling mechanistic studies of nucleic acid encapsulation, pH-dependent structural transitions, membrane fusion, and endosomal escape [5,6]. These simulations reveal how membrane phase behavior governs the assembly, structure, and delivery efficiency of LNP formulations. Finally, I will briefly discuss how combining CG simulations with machine learning can accelerate the rational design of next-generation lipid nanoparticles.

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